

Part I: Objective (20 points)

For the following questions select the best of the given options and write your choice on the answer sheet provided at the end of Part I. All questions have equal points.

1. Genetic algorithm halts when:

- ☒ (A) Fitness threshold is not achieved but the specified generations expire
- ☐ (B) Crossover can't be applied to chromosome(s)
- ☐ (C) Mutation generates clones of chromosome(s)
- ☐ (D) All chromosomes fall below the fitness threshold

2. Roulette wheel selection in genetic algorithms can eliminate even the fittest solutions.

- ☒ (A) True
- ☐ (B) False

3. Mutation helps transfers fitness characteristic into the next generation of candidate solutions.

- ☐ (A) True
- ☒ (B) False

4. Genetic algorithm terminates when:

- ☐ (A) Specified generations expire
- ☐ (B) Required fitness threshold is achieved
- ☒ (C) Both 'A' and 'B'
- ☐ (D) None of the given options

5. In unifying predicate calculus expressions, if a variable is bound to a constant, it may not be given a new binding in future unification.

- ☒ (A) True
- ☐ (B) False

6. In first order predicate calculus, quantified variables refer to relations in the problem domain.

- ☐ (A) True
- ☒ (B) False

7. The expression $\exists X (p(X) \vee \neg p(X))$ is not consistent.

- ☐ (A) True
- ☒ (B) False

8. Logic based automated reasoners are an example of strong method problem solving in AI.

- ☒ (A) True
- ☐ (B) False

9. Composition of unification bindings is commutative but not associative.

- ☐ (A) True
- ☒ (B) False

10. Clause form has all variables _____ quantified.

- ☒ (A) Universally
- ☐ (B) Existentially

11. An inference rule is _____ if every predicate calculus expression produced by it logically follows from the given set of predicate calculus expressions.

- ☒ (A) Complete
- ☐ (B) Sound
- ☐ (C) None of the given options

12. _____ check ensures that a variable is not unified with a term containing it.

- ☒ (A) Unification
- ☐ (B) Model
- ☐ (C) Logically follows
- ☐ (D) None of the given options

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13. Horn clauses have at _____ one positive literal.

- (A) Least (B) Most

14. Information gain metric in decision tree induction measures:

- (A) Gain in information of the dataset
(B) Reduction in entropy due to partitioning by an attribute
(C) Entropy of an attribute

(D) Gain in entropy of the dataset

15. An attribute in a categorical dataset that has many values, e.g., Date, has:

- (A) Low entropy and high gain
(B) High entropy and low gain
(C) Low entropy and low gain
(D) High entropy and high gain

16. Maximum entropy of a dataset having three classes is:

- (A) 0.5 (B) 1.25 (C) 0.75 (D) 1.58

17. In decision tree induction, more split information of an attribute means more gain ratio.

- (A) True (B) False

18. In ID3 algorithm, if for some attribute entropy is maximum then it would be selected as the node.

- (A) True (B) False

19. An attribute may be present in more than one branch of a decision tree.

- (A) True (B) False

20. If all samples of a dataset belong to only one class, ID3 would produce a tree with _____ nodes.

- (A) 2 (B) 0 (C) 1 (D) None of the given options

Part II: Subjective (60 points)

Note: Answer all questions. Limit your responses to the space provided. Maximum points for each part are given next to the question number in square brackets.

Q1. [2+8+2]

(a) Consider bit-level representation for traveling salesperson problem involving 'n' cities. Describe two issues in designing genetic operators for this representation.

In bit-level representation, when we apply genetic operators, even changing 1-bit could cause drastic change to our candidate solution. The result could go out of our selected range. Our minima could turn into an extrema & vice versa. In simple word, good solution can become worse & vice versa.

(b) For the following candidate solutions, apply order-crossover as discussed in the class to create two new solutions c1 and c2. You must take the cut points as follows: first between city index 3 and city index 4, second between city index 7 and city index 8.

$$p1 = (1, 9, 2, 4, 6, 5, 7, 8, 3)$$

$$p2 = (4, 5, 9, 1, 8, 7, 6, 2, 3)$$

$$\begin{matrix} 1 & 9 & 2 & 4 & 6 & 5 & 7 & 8 & 3 \\ 4 & 5 & 9 & 1 & 8 & 7 & 6 & 2 & 3 \end{matrix}$$

C1: $[6, 5, 7, 8]$
take right part from p2, place at left

C1: $2, 3, [6, 5, 7, 8]$
take left part from p2, place at right

$$2, 3, [6, 5, 7, 8], 9, 1, [8]$$

Result

$$C1 = (2, 3, 4, 6, 5, 7, 9, 1, 8)$$

(c) Give two mutation operators for the above problem.

- 1:- 2-point cross-over
- 2:- Multiple-point cross-over

C2: $[1, 8, 7, 6]$
take left part of p1, place at left

C2: $3, [1, 8, 7, 6]$
take left part of p1 place at right

$$3, [1, 8, 7, 6], 9, 2, [4, 5]$$

Result

$$C2 = (3, 1, 8, 7, 6, 9, 2, 4, 5)$$

$$p \vee \neg p \quad (p \vee \neg p) \vee (p \vee \neg p)$$

Q.2. [5+6+4]

(a) Convert the following into first order predicate calculus using the syntax and conventions discussed in the class:

i) Busy people sometimes have friends

$$\forall x, \text{busy}(x) \rightarrow \text{sometimes have}(\text{friends}, x)$$

ii) All romans either liked Caesar or hated him. [Note: take 'or' in exclusive sense]

$$\forall x \text{ romans}(x) \rightarrow (\neg \text{liked}(x, \text{caesar}) \wedge \text{hated}(x, \text{caesar})) \vee (\text{liked}(x, \text{caesar}) \wedge \neg \text{hated}(x, \text{caesar}))$$

iii) Any good amateur can beat some professional

$$\forall x \text{ can beat}(\text{professional}, \text{good amateur}, x) \vee x \text{ good amateur}(x) \rightarrow \text{can beat}(\text{profess}, x)$$

iv) No one sees everything

$$\forall x \neg \text{sees everything}(x)$$

v) All students of a philosopher read one of their teacher's books

$$\forall x \text{ student of}(\text{philosopher}, x) \rightarrow \text{read book of teacher}(x)$$

(b) Convert the following sentences into clause form:

i) $\forall x p(x) \rightarrow (a(x) \wedge b(x)) \vee \neg c(x, d)$

ii) $\exists y q(y, c) \wedge (\forall z a(z) \rightarrow \neg b(y))$

$$i) \forall x p(x) \rightarrow (a(x) \wedge b(x)) \vee \neg c(x, d)$$

Step 1: Remove implications:-

$$\forall x \neg p(x) \vee (a(x) \wedge b(x)) \vee \neg c(x, d)$$

Step 2: Drop quantifiers:-

$$\neg p(x) \vee (a(x) \wedge b(x)) \vee \neg c(x, d)$$

Step 3: Conjunction of disjunction form

$$\neg p(x) \vee a(x) \wedge \neg p(x) \vee b(x) \vee \neg c(x, d)$$

Final form:-

$$a_x = \neg p(x) \vee a(x) \\ b_x = \neg p(x) \vee b(x) \vee \neg c(x, d) \quad \text{Ans}$$

$$\begin{aligned}
 E(\text{Fair}) &= -(P_{\text{good}} \log_2 P_{\text{good}}) - (P_{\text{ungood}} \log_2 P_{\text{ungood}}) \\
 &= -\left(\frac{0}{86} \log_2 \frac{0}{86}\right) - \left(\frac{66}{86} \log_2 \frac{66}{86}\right) = -\left(\frac{66}{86} \log_2 \frac{66}{86}\right) \\
 &= 0 \\
 &= 0.52 + 0.47 = 0.99
 \end{aligned}$$

$$\begin{aligned}
 I(\text{Gender}) &= 0.98 - 0.45 - 0.52 \\
 &= 0.98 - 0.45 - 0.52 = 0.01 \quad \text{Ans}
 \end{aligned}$$

$$\begin{aligned}
 I(\text{GPA}) &= 0.98 - \frac{23}{228}(0) - \frac{65}{228}(0.96) - \frac{74}{228}(0.9) - 0 \\
 &= 0.98 - 0.27 - 0.29 = 0.42 \quad \text{Ans}
 \end{aligned}$$

(d) If unique record identifiers are taken as regular attributes while inducing a decision tree from data, they tend to become the root node due to high 'Gain'. What could be an alternative to using 'Gain' as a metric for decision tree induction?

'Gain' method can't be taken in case of unique record identifiers taken as regular attributes because their gain would be too high which will result in being selected as root.

To solve this problem, we use

'Gain Ratio'. By using gain ratio as metric for decision tree, we can ~~split~~ those unique identifiers so they won't be selected. Ratio gain will become low as we split it.

$$ii) \exists Y q(Y, c) \wedge (\forall Z a(z) \rightarrow \neg b(Y))$$

Step 3:-

$$\exists Y q(Y, c) \wedge (\forall Z \neg a(z) \vee \neg b(Y))$$

Step 4:-

$$\exists Y \forall Z q(Y, c) \wedge (\neg a(z) \vee \neg b(Y))$$

Step 5:-

$$\forall Z q(f(d), c) \wedge (\neg a(z) \vee \neg b(s(d)))$$

use Skolem const.

Step 6:-

$$q(f(d), c) \wedge (\neg a(z) \vee \neg b(s(d)))$$

Step 7:-

$$\neg b(s(d)) \vee \neg a(z) \quad \neg b(s(d)) \vee q(f(d), c)$$

final form:-

$$ax: \neg b(s(d)) \vee \neg a(z)$$

$$bx: \neg b(s(d)) \vee q(f(d), c)$$

} Ans

(c) Given the following set of clauses including the goal clause, use resolution proof procedure with unit preference strategy to show whether the goal is true or not.

1. $\text{poor}(X) \vee \neg \text{smart}(X) \vee \text{happy}(X)$	4. $\neg \text{poor}(\text{john})$
2. $\neg \text{read}(Y) \vee \text{smart}(Y)$	5. $\neg \text{happy}(Z) \vee \text{exciting}(Z)$
3. $\text{read}(\text{john})$	6. $\neg \text{exciting}(W)$ (Negated Goal)

$$\neg \text{exciting}(w) \quad \neg \text{happy}(z) \vee \text{exciting}(z)$$

$$\{z/w\}$$

$$\neg \text{happy}(z) \quad \text{poor}(x) \vee \neg \text{smart}(x) \vee \text{happy}(x)$$

$$\{x/z\}$$

$$\text{poor}(x) \vee \neg \text{smart}(x)$$

$$\neg \text{poor}(\text{john})$$

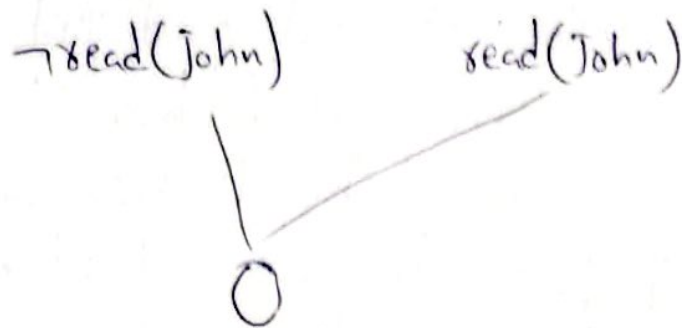
$$\{\text{john}/x\}$$

$$\neg \text{smart}(\text{john})$$

$$\neg \text{read}(y) \vee \text{smart}(y)$$

$$\{\text{john}/y\}$$

$$\neg \text{read}(\text{john})$$



Since, we got null

It means our goal is true.

(9)

Q.3. [5+4+3+3]

(a) Give any five rules of deductive inference.

- 1:- Modus Ponens

if $P \rightarrow Q (T)$	{	3:- And introduction	{	5:- <u>?</u>
$P (T)$		if $a (T)$		if the whole domain
Then $Q (T)$		$b (T)$		is true, then sub part
		then $a \wedge b (T)$		of that domain is
- 2:- Modus Tollens

if $P \rightarrow Q (T)$	{	4:- And elimination	{	if the whole domain
$\neg Q (T)$		if $a \wedge b (T)$		is true, then sub part
then $P (T)$		$a (T)$		of that domain is
		then $b (T)$		also true.

(b) How does Prolog differ from imperative programming languages, like C/C++? How does it prove a theorem?

Prolog uses clauses in a database and apply resolution on them to find if theorem is true or false.

It takes negation in the database. Since whole database the negation will disrupt the database null on applying resolution. If it we say that since negative is false, must be true. 2

(c) What is Horn Normal Form? Explain.

HNF stands for Horn Normal Form. It contains one positive literal. HNF ~~is~~ is ^{clauses} disjunctive. It can be converted into CNF. 15

(d) Differentiate between unit resolution and the unit preference strategy.

Unit Preference:-

This strategy says that, if there is unit clause present in database, then it is that we resolve it first. It can be present as any clause. 0.5

Unit resolution:-

There is no necessary condition implemented that we must have to resolve unit clause we resolve any clause from database with parent and we continue resolution as normal. It is a practice to resolve them first but not necessary in this strategy.

Q.4. [5+2+8+5]

19

(a) What is the information content, as measured by the Entropy, of a fair coin? How would it change if the coin has been rigged to come up heads $\frac{1}{4}$ of the time?

Entropy of a fair coin would be 1. Because there is $\frac{1}{2}$ chance of both head & tail.

$$E(s) = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$$

If coin is rigged, then it means heads has $\frac{3}{4}$ chance while tail has $\frac{1}{4}$

$$E(s) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.31 + 0.5 = 0.81$$

Information content is the measure of reduction in entropy.

(b) What is the inductive bias of ID3 algorithm?

Inductive bias of ID3 algorithm is that if we take unique record identifiers they have high gain and become root node, which is not feasible. So ID3 slices that attribute so, it won't become root. One way to do it is ratio gain instead of gain.

(c) We are given a task to build a decision tree classifier for the following representative sample of students' data (which has been condensed in two tables for the two classes of students: graduate and undergraduate.) Calculate information gain for the attribute 'GPA'.

Note that 'Count' is NOT a regular attribute; it just gives the frequency of occurrence of the specific record.

Class: Graduate Students

Gender	Major	Age Range	GPA	Nationality	Count
M	Science	21...25	good	Pakistani	16
F	Science	21...25	excellent	Pakistani	22
M	Engineering	26...30	excellent	Foreigner	18
F	Science	26...30	very good	Foreigner	25
M	Engineering	26...30	excellent	Pakistani	10
F	Engineering	21...25	good	Pakistani	07

good = 23
exl = 50
V-good = 25
fair = 0